



Unit 4 · Outcome 1 · Climate Change

What tips Earth's energy balance?

KK01

VCE Environmental Science · Units 3 & 4 · Exam study summary

Digital Vector EnviroSci · single-topic mastery summary — clean explanations, comparison tables, visual aids, exam traps, command words and worked answers.



CORE IDEA

Earth's energy balance

Almost all of Earth's energy arrives as incoming solar radiation — high-energy shortwave light from the Sun. It leaves again in two ways: some is reflected straight back to space without ever warming anything, and the rest is absorbed, turned into heat, then radiated away as lower-energy longwave (infrared) radiation. Earth's energy balance is simply the comparison of energy in against energy out at the top of the atmosphere.

When energy in = energy out, the planet is in equilibrium and its average temperature stays steady. If more comes in than leaves, the surplus heat builds up and Earth warms. If more leaves than arrives, Earth cools. That mismatch has a name you'll meet in the news and the exam: an energy imbalance.

◆ ANALOGY

The bathtub. Picture a bath with the tap running (energy in) and the plug half-out (energy out). If inflow matches outflow the water level holds. Open the tap or jam the plug and the level rises; that rising water is heat accumulating in the climate system. The temperature isn't set by the tap alone — it's set by the difference between tap and drain.

Crucially, the Sun's raw output isn't the only thing in the equation. Anything that changes how much sunlight gets in, or how easily heat gets out, shifts the balance — even if the Sun itself never changes. That is exactly how a volcano or a thicker blanket of greenhouse gas can move the thermometer.

THE LINK WORD

Radiative forcing: tipping the scales

A radiative forcing is any factor that pushes the energy balance away from equilibrium. The direction matters and the exam rewards you for naming it:

- Positive forcing → warming. The factor adds energy to the system — either letting more sunlight in or stopping heat escaping (e.g. more greenhouse gas).
- Negative forcing → cooling. The factor removes energy — reflecting sunlight away before it's absorbed (e.g. volcanic haze).

◆ ANALOGY

A thumb on the scales. Think of energy-in and energy-out as two pans of a balance that normally sit level. A forcing is a thumb pressing on one side. Press the "in" side (more sun, less escape) and the warm pan drops. Press the "out" side (reflect sunlight away) and the cool pan drops. The three KK01 factors are simply three different thumbs.

Keep two more labels straight, because VCAA loves to test them together with direction:

- Natural vs anthropogenic — did nature do it (volcano, Sun) or did humans (changing the air's composition)?
- Timescale — does the push last a couple of years (volcanic haze), swing over decades (sunspot cycle), grind over tens of thousands of years (orbital cycles), or keep building for centuries (our emissions)?

NATURAL · COOLING

Factor 1 — Volcanic eruptions

A big explosive eruption blasts sulfur dioxide high into the stratosphere, where it forms a haze of tiny sulfate aerosol droplets. That haze acts like a planet-wide sunshade: it reflects incoming sunlight back to space before it can be absorbed, so less energy enters the system and Earth's surface cools. This is a negative radiative forcing — and a natural one.

The Philippines eruption lofted ~20 million tonnes of SO₂ into the stratosphere. Global average temperatures fell by roughly 0.5 °C for about two years before the haze settled out and warming resumed. The classic VCAA example of a short, sharp natural cooling.

The largest eruption in recorded history dimmed skies worldwide. 1816 became "the Year Without a Summer": crops failed and snow fell in June across parts of Europe and North America. A vivid reminder that the effect is real but temporary.

⚠ WATCH OUT

The CO₂ trap. Volcanoes do release carbon dioxide, but the amount is tiny next to human emissions, and the short-term aerosol cooling dominates. So the exam-correct headline is: a major eruption cools the planet for a year or two. Don't let "volcanoes make CO₂" tempt you into writing "volcanoes cause warming."

NATURAL · EITHER WAY

Factor 2 — Solar variability

The Sun is the tap, so changes in its output change the whole budget. Solar variability works on two very different clocks:

The number of sunspots rises and falls about every 11 years, nudging total solar output up and down by roughly 0.1%. More active Sun → a touch more incoming energy → slight warming; quiet Sun → slight cooling. The Maunder Minimum (a 17th-century lull in sunspots) coincided with the cold "Little Ice Age."

Over very long timescales, slow wobbles in Earth's orbit change how much and where sunlight lands: eccentricity (shape of the orbit, ~100,000 yr), obliquity (axial tilt, ~41,000 yr) and precession (axial wobble, ~26,000 yr). Together they pace the glacial–interglacial cycle — the natural rhythm of the ice ages.

◆ NOTE

Why this matters for the exam. Solar variability is the go-to natural explanation for past climate swings. But its modern effect is small and has no upward trend — measured sunlight has been flat-to-slightly-down since 1980 while temperatures climbed. That mismatch is precisely why scientists point to the third factor for recent warming.

ANTHROPOGENIC · WARMING

Factor 3 — Human change to atmospheric gases

The third factor is the only anthropogenic one, and it's the dominant driver of the warming measured at Cape Mornay this century. Human activities are changing the gas composition of the atmosphere — raising the concentration of greenhouse gases like carbon dioxide (CO₂), methane (CH₄) and nitrous oxide (N₂O):

- Burning fossil fuels (coal, oil, gas) for electricity, transport and industry — the largest source of extra CO₂.

- Land-use change & deforestation — clearing forests releases stored carbon and removes a sink.
- Agriculture — livestock and rice paddies release methane; fertilisers release nitrous oxide.
- Cement production — chemically releases CO₂ as limestone is processed.

Extra greenhouse gas doesn't block the incoming sunlight — it makes it harder for heat to escape. More outgoing infrared is absorbed and re-radiated back down, so energy leaves more slowly than it arrives. The result is a positive radiative forcing → warming. (The mechanism itself — the greenhouse effect — is the focus of KK02 and KK04; here you just need it as one of the three factors and its direction.)

◆ ANALOGY

A thicker doona. Greenhouse gases are like a blanket over the bed. The blanket doesn't make you produce more heat — it just slows the heat you already give off from escaping, so you warm up under it. Add layers (more gas) and you get warmer still. Unlike a volcano's haze, this blanket isn't settling out after two years — we keep adding to it, so the push is sustained and building.

INTERACTIVE

The Cape Mornay energy desk

Here's the station's control desk. Flip each factor and watch the balance beam tip and the thermometer respond. Try to predict the direction before you toggle — that prediction is exactly the exam skill.

• NOTE

Watch for this: turn on the volcano and greenhouse gas together. They push in opposite directions, but the haze settles out in a couple of years while the gas keeps building — which is why a volcano can only ever pause warming, never reverse it.

TAP TO REVEAL

Decode the command word

◆ COMMAND WORDS

Marks are lost when students answer the wrong kind of question. The command word tells you exactly what the examiner wants. Tap each one.

Command word	What the marker wants
Identify / State	Name it — no explanation.
Describe	Give the features / what happens, in order.
Explain	Give the how / why — cause and effect.
Distinguish / Compare	State the difference (and similarity) explicitly.
Analyse	Break into parts and show how they relate.
Evaluate / Justify	Weigh strengths & limits, then a reasoned verdict.

Match your answer to the verb — the fastest marks in the exam.

AVOID THESE

Six exam traps

⚠ EXAM TRAPS

The dominant short-term effect is cooling from reflective sulfate aerosols. Volcanic CO₂ is trivial next to human emissions. Write: eruptions cause short-term cooling. Earth's temperature is set by the difference between energy in and out, not by inflow alone. Always frame it as a balance / imbalance. They don't stop sunlight coming in — they slow outgoing infrared from leaving. Specify it's the heat escaping that's reduced. Solar output has been flat-to-slightly-falling since ~1980 while temperatures rose. The trends don't match — name human gas change as the driver. "It causes warming" earns little. Show the chain: factor → change to energy in/out → warming or cooling. Volcanoes and solar variability are natural; only the gas-composition change is anthropogenic. Label each one correctly.

LIVE FEEDBACK

P·E·L writing trainer



The P·E·L answer shape — make a Point, support with Evidence, then Link to the question.

🔑 P·E·L WRITING

High-scoring answers have a shape. Build the habit here: make a Point, back it with Evidence, then Link it to the energy balance. Type below and the meters light up as you hit each part.

-MARK QUESTION

03 Good answer vs bad answer

Question: Explain how human changes to the atmosphere's gas composition affect Earth's energy balance. (3 marks)

Humans put pollution in the air and this makes the Earth hotter because of global warming. The greenhouse gases block the sun so it gets warmer.

Human activities such as burning fossil fuels raise the concentration of greenhouse gases like CO₂. These gases absorb and re-radiate outgoing infrared radiation, so heat escapes to space more slowly. Less energy leaves than enters, creating a positive imbalance, so Earth's surface warms.

MARKS · SELF-MARK

22 Exam-style questions

★ PRACTICE & SELF-MARK

Write your answer first, then reveal. Use the marking guide to score yourself honestly and tap your mark — the dial in the corner keeps your running total.

BEFORE YOU CLOSE THE TAB

Quick recap

- Energy balance = incoming solar (shortwave) vs reflected + outgoing infrared (longwave). In = out → stable temperature.
- Radiative forcing tips the balance: positive → warming, negative → cooling.
- Volcanic eruptions — natural, short-term cooling: stratospheric sulfate aerosols reflect sunlight away (Pinatubo 1991, Tambora 1815).
- Solar variability — natural, either direction: ~11-yr sunspot cycle (small) and Milankovitch orbital cycles (ice ages). Modern trend is flat — can't explain recent warming.
- Human gas-composition change — anthropogenic, sustained warming: more greenhouse gas slows outgoing infrared, so energy out < in.
- In an "explain": factor → change to energy in/out → warming or cooling. Always close the loop on the balance.

EXAM TECHNIQUE

★ Worked answers — weak vs strong

The same question answered two ways. The strong column shows the moves that earn the marks.



The P-E-L answer shape — make a Point, support with Evidence, then Link to the question.

✗ WEAK ANSWER

Humans put pollution in the air and this makes the Earth hotter because of global warming. The greenhouse gases block the sun so it gets warmer. Why only 1 mark: vague ("pollution"), circular ("hotter because of global warming"), and a factual error — gases don't block the Sun coming in. No mechanism, no link to the balance.

✓ STRONG / MODEL ANSWER

Human activities such as burning fossil fuels raise the concentration of greenhouse gases like CO₂. These gases absorb and re-radiate outgoing infrared radiation, so heat escapes to space more slowly. Less energy leaves than enters, creating a positive imbalance, so Earth's surface warms. Mark-earning moves: named human source + named gas (P), correct mechanism on outgoing infrared (E), and an explicit tie to energy out < energy in → warming (L).